DIY Activity: Analyzing Recall Patterns & Product Insights

Guidelines for Students: *This is a self-paced Do-It-Yourself (DIY) activity and* ***submissions are not required****. The primary objective is to enhance your skills through practical, hands-on learning. Engage in discussions with peers to deepen your understanding and collaboratively work through the activity. If you have any questions or need clarification,* ***please raise them during the upcoming Live Sessions*** *for a more comprehensive explanation.*

Overview: *A quality assurance team is analyzing product recall data for a large organization. They need to explore the data using MS Excel's statistical, logical, text, and date functions. The analysis will help the team understand trends, make decisions, and generate reports efficiently. Additionally, AI tools like Flash Fill and Recommended Charts will be leveraged for advanced insights.*

Dataset:

product_recall Dataset

- ***Product ID-*** *Unique identifier for each product.*
- ***Batch Number-*** *Identifier for the production batch of the product.*
- ***Manufacturing Date-*** *Date the product was manufactured.*
- ***Recall Date-*** *Date the product was recalled. Recall is defined as the product retrieval due to defects*
- ***Defect Rate-*** *Proportion indicating the rate of defects in the product. To convert it to a percentage, you would multiply the value by 100.*
- ***Units Recalled-****The total number of units recalled for the product.*
- ***Reason for Recall-****The reason specified for the product's recall.*

Tasks to be performed:

- *The team wants to find how many recalls had a 'Defect Rate' greater than 15%. Use a STATISTICAL function to count the number of such entries. What insights can the team derive from the count?*

- *Apply CONDITIONAL FORMATTING to the 'Recall Date' column based on the following criteria: format the cells in green for recalls that occurred within six months of the manufacturing date, in yellow for recalls that occurred within one year but more than six months after the manufacturing date, and in red for recalls that occurred more than one year after the manufacturing date. After applying the formatting, determine how many recalls fall into each category.*

- *Calculate the total number of Units Recalled for all recalls where the Reason for Recall is 'Defective Product.' How does this contribute to the analysis of defect trends by indicating what percentage this count represents of the total units recalled?*

- *Determine the average Defect Rate for recalls made in 2023. What does this tell the team about the quality control issues in the specified year in relation to the total products manufactured?*

- *Identify the maximum and minimum Units Recalled across all entries. Which batch numbers are associated with these recalls, and what are the reasons for recall?*

- *The team wants to classify recalls into two categories using a logical function: "High Risk" if the Defect Rate is greater than 20% or if the Units Recalled exceed 500, and "Low Risk" otherwise. Further determine how many recalls fall into each category based on the results of the classification using the COUNT function.*

- *Identify recalls where the 'Reason for Recall' contains "Regulation Violation" AND the 'Defect Rate' is greater than 10%. Highlight these recalls using a logical function.*

- *Determine the total number of recalls where the 'Units Recalled' are less than 50 OR the 'Defect Rate' is less than 5%. What does this indicate about minor issues in the dataset?*

- *Create a unique identifier for each product by combining the 'Product ID', 'Batch Number', and 'Reason for Recall' into a single string, separated by hyphens by using CONCATENATE function. What is the identifier for the first recall in the dataset?*

- *Extract the first three characters of the 'Batch Number' using the LEFT function to determine the batch series for each product. How many unique batch series exist in the dataset using the COUNTA function?*

- *Use the LEN function to calculate the length of each 'Reason for Recall'. Which reason has the maximum length, and what does this indicate about its complexity?*

- *Calculate the number of days between the 'Manufacturing Date' and 'Recall Date' for each entry with the help of an ARITHMETIC operator. What is the average time taken for recalls across all products?*

- *Identify recalls that occurred in the same year as the 'Manufacturing Date' by calculating the YEAR of both dates and comparing them. How many such recalls exist, and what could this imply about immediate quality control checks?*

- *Use a DATE function to format January 1, 2023, and find all recalls made after this date. How many products fall into this category?*

- *Use the ANALYZE DATA feature in MS Excel to identify patterns in the 'Defect Rate'. Which patterns emerge, and how can they assist in decision-making for quality improvements?*

- *Use the RECOMMENDED CHARTS feature to visualize the relationship between 'Defect Rate' and 'Units Recalled'. What is the most suitable chart type, and what insights does it provide?*

- *Using the FLASH FILL feature, format the 'Manufacturing Date' column into "MM-YYYY" format. How many products were manufactured in the same month as their recall?*
  (Hint: Use the TEXT function to format the Manufacturing Date)

- *Leverage FLAS FILL to extract the numeric portion of 'Batch Number'. How many unique numeric batch identifiers exist, and what do they reveal about production trends?*

- *Use conditional formatting to highlight rows where the Units Recalled are in the top 10% of the dataset. What corrective actions can the team take based on these highlights?*